

CHEMICAL THERMODYNAMICS

Thermodynamics is the branch of science which deals with the conversion of heat into mechanical work and vice versa. In chemical thermodynamics, we confine our study to chemical substances and chemical changes. In Engineering the use of thermodynamics lies in its ability to predict the maximum efficiency of various types of heat engines and maximum work obtainable from a given fuel. It also helps to predict the feasibility of a physical or chemical process under a given set of conditions.

Thermodynamic terms

System

It can be defined as the part of the universe which is selected for study and is separated from the rest of the universe by definite boundary

Types of systems

Depending on the interaction between the system and surroundings, there are three different types of systems.

1. **Open systems:** Systems which can exchange both matter and energy with the surrounding. Eg. Hot liquid taken in a beaker.
2. **Closed systems:** Systems which can exchange energy but not the matter with the surrounding. Eg. Hot liquid taken in a closed metallic flask.
3. **Isolated systems:** Systems which can exchange neither energy nor matter with the surrounding. Eg. Hot liquid taken in a thermally insulated closed flask.

Surrounding: Rest of the universe other than the system is known as surroundings

Properties of a system

- a) **Extensive property:** Properties which depend upon the amount of substances present in the system. eg. Volume, mass, internal energy, free energy, enthalpy, entropy, etc.
- b) **Intensive property:** Properties that do not depend upon the amount of substances present in the system. Eg. Density, molar entropy, molar heat capacity, etc.

State variables: The measurable properties of a system which completely define the state of a system are called state variables or thermodynamic variables. Eg. Pressure, volume, temperature, concentration, etc.

State of a system: A thermodynamic system is said to be in a definite state when the state variables have definite values. Any change in the value of these variables will change the state of the system.

Process: A thermodynamic process is defined as the method of operation with the help of which a system changes from one state to another. The various thermodynamic processes are:

a) **Isothermal process:** It is the process carried out at constant temperature

b) **Adiabatic process:** A process carried out in a system such that there is no heat exchange between system and surroundings. In adiabatic process temperature of the system can change.

Reversible process: A process in which the driving force is only infinitesimally greater than the opposing force at any instant such that the process can be reversed in direction by making a small change in any property of the system, such as pressure, temperature, etc.

A truly reversible process cannot be realized in practice.

Irreversible process: It is a process in which the system or surroundings are not restored to their initial states at the end of the process. All processes occurring spontaneously in nature are irreversible. Eg. Diffusion of gases, flow of heat from a hotter body to a cooler body, etc.

Internal energy: The total energy contained in a thermodynamic system is called internal energy. It includes all forms of kinetic, as well as potential energy. Kinetic energy is associated with all kinds of motions, such as translational, rotational and vibrational, of the system's particles. Potential energy is associated with the static energy of the particles. Internal energy is a function of the temperature of the system. Since the value of internal energy depends on the amount of matter in the system, it is an extensive property. It is a state function since its value depends on the state of the system and not on the path followed to obtain it. Internal energy is represented by E .

The molecular interpretation of internal energy is the concept of the energy being shared or partitioned equally over all modes of motion of a molecule. Thus, rotation around an axis has the same average energy as translation along a coordinate, and so on.

Heat capacity of a system

It is defined as the heat (q) required to raise the temperature of the system from the lower temperature (T_1) to the higher temperature (T_2) divided by the temperature difference ($T_2 - T_1$). Mathematically, $C = \frac{q}{(T_2 - T_1)} = \frac{q}{\Delta T}$

Thus, heat capacity of a system is the quantity of heat required to raise the temperature of the whole mass of it through 1°C .

Specific heat is the quantity of heat required to raise the temperature of unit mass of it through 1°C .

Sign convention

The energy transferred as heat to a system from its surroundings is taken as positive while the energy transferred as heat from the system to its surroundings is taken as negative.

Similarly, the work done on the system by its surroundings is taken as positive while the work done by the system on its surroundings is taken as negative.

First law of thermodynamics

It is basically the law of conservation of energy which states that energy can neither be created nor destroyed although it can be converted from one form to another. This law holds good in chemical reactions also. Energy can be absorbed or evolved in a chemical reaction but the total energy of the reacting system and the surrounding remain constant.

Mathematical expression of the first law of thermodynamics

Let us consider a system whose internal energy is E_1 . Now, if the system absorbs q amount of heat, and if work(w) is done on the system, then the internal energy of the system increases and becomes $E_1 + q + w$, say E_2 .

$$\text{Thus, } E_2 = E_1 + q + w$$

$$\text{Or } E_2 - E_1 = q + w \quad \text{or } \Delta E = q + w$$

i.e., Change in internal energy = Heat added to the system + work done on the system

If the system does work on the surroundings, its internal energy decreases. In this case, work is taken as negative ($-w$).

$$\Delta E = q + (-w) = q - w$$

i.e., Change in internal energy = Heat added to the system - work done by the system

If a process is carried out at constant temperature (isothermal process), the change in internal energy is zero.

$$\text{i.e., } \Delta E = q - w = 0 \quad \text{or } q = w$$

If a reaction is carried out at constant volume, then no work is done and the heat exchanged with the surroundings is equal to change in internal energy

$$\Delta E = q + 0 = q$$

For a process carried out at constant pressure (i.e., in an open vessel),

$$Q = \Delta E + w$$

i.e., Amount of heat given to a system is partly used to increase the internal energy of the system and partly used for doing work done by the system.

If only pressure-volume work is done and the pressure is constant equal to the atmospheric pressure,

$$\text{Then, } w = P (\Delta V) = P(V_2 - V_1)$$

$$\text{i.e., } q_p = \Delta E + P (\Delta V) = \Delta E + P(V_2 - V_1)$$

$$= (E_2 - E_1) + P(V_2 - V_1) = (E_2 + PV_2) - (E_1 + PV_1) = (E + PV)_2 - (E + PV)_1$$

Since heat absorbed at constant pressure is equal to difference of two quantities that are functions of the states of the system, it is convenient to introduce a new state function, the enthalpy, H , which is defined by $H = E + PV$

Thus, $q_p = H_2 - H_1 = \Delta H = \Delta E + P(\Delta V)$

The heat absorbed in a process at constant pressure is equal to the change in enthalpy if the only work done is reversible pressure-volume works.

For solids and liquids, ΔV is very small; therefore, $P(\Delta V)$ is insignificant and $\Delta H = \Delta E$.

For gases, ΔV and hence $P(\Delta V)$ have significant values, therefore at constant pressure, $\Delta H > \Delta E$.

In Chemical reactions involving gases,

$$\Delta H = \Delta E + P(\Delta V) = \Delta E + P(V_P - V_R)$$

From ideal gas laws, $PV_R = n_R RT$ and $PV_P = n_P RT$

$\Delta H = \Delta E + (n_P RT - n_R RT) = \Delta E + (n_P - n_R) RT = \Delta E + \Delta n RT$ where Δn is the difference between the number of moles of gaseous products and gaseous reactants.

Work done in a reversible isothermal expansion/compression of an ideal gas

Suppose a gas undergoes reversible isothermal expansion from V_i to V_f .

From the definition of work, $w = - \int_{V_i}^{V_f} P_{ext.} dV$

For a reversible process, $P_{ext.} = P$ of the gas. Hence $w = - \int_{V_i}^{V_f} P dV$

For an ideal gas, we have $PV = nRT$

Or $P = nRT/V$

$$w = - \int_{V_i}^{V_f} \frac{nRT dV}{V}$$

For an isothermal process,

$$W = - nRT \int_{V_i}^{V_f} \frac{dV}{V}$$

$$\text{i.e., } w = nRT \ln \frac{V_i}{V_f} = nRT \ln \frac{P_f}{P_i}$$

Entropy

Entropy is a measure of the randomness or disorder of the molecules of the system. It is denoted by S . For a spontaneous process $S_f > S_i$ or $\Delta S > 0$

For a reversible change taking place at a fixed temperature (T) the change in entropy (ΔS) is equal to heat energy absorbed or evolved divided by the temperature (T)

$$\text{i.e., } \Delta S = \frac{q_{rev}}{T}$$

Entropy is expressed in units which in SI system corresponds to $\text{J mol}^{-1}\text{K}^{-1}$ and in cgs system $\text{cal mol}^{-1}\text{K}^{-1}$

Gibb's Free energy function

It is defined as $G = H - TS$ where G is Gibb's free energy, H enthalpy, T temperature and S entropy.

Gibb's free energy is a state function. Hence, for a cyclic process, $\Delta G = 0$

It is an extensive property.

Significance of Gibb's free energy

1. It is used for predicting the spontaneity of a process under different conditions.

To find the criterion of spontaneity in terms of G , we first write down differential of

$$G = H - TS$$

$$dG = dH - TdS - SdT$$

When pressure and temperature are constants,

$$dT = 0,$$

$$dH = dq,$$

$$dG = dq - TdS$$

But, by definition of entropy, $TdS = dq_{rev}$.

Now, there are two possibilities. If a process is reversible, $dq = dq_{rev}$, so

$$dG = 0 \text{ (reversible process, constant P,T)}$$

If a process is irreversible, $q < q_{rev}$, and $dq < dq_{rev}$.

$$dG = dq - dq_{rev}$$

$$dG < 0 \text{ (irreversible process, constant P,T)}$$

These equations apply to infinitesimal changes. For finite changes, they become

$$\Delta G = 0 \text{ (reversible process, constant P,T)}$$

$$\Delta G < 0 \text{ (irreversible process, constant P,T)}$$

To decide whether a process will be spontaneous when carried out at constant temperature and pressure, we have only to calculate ΔG for the system alone. If $\Delta G < 0$, the process will be spontaneous. If $\Delta G = 0$, the initial and final state can exist in equilibrium with each other, with no net change. If $\Delta G > 0$, the process will not occur spontaneously, but the reverse process will.

2. The decrease in free energy is equal to the amount of maximum work which a system can do at constant temperature and pressure over and above the mechanical work.

$$G = H - TS$$

$$G = E + PV - TS$$

$$dG = dE + PdV + VdP - TdS - SdT$$

But, by definition of entropy, $TdS = Dq = dE + W_{rev}$.

$$\text{Or } dE = TdS - W_{rev}$$

$$\text{i.e., } dG = TdS - W_{rev} + PdV + VdP - TdS - SdT = -W_{rev} + PdV + VdP - SdT$$

Now, W_{rev} = mechanical work + non mechanical work

$$= PdV + W_{net}$$

$$\text{Therefore, } dG = -PdV - W_{net} + PdV + VdP - SdT$$

$$= VdP - SdT - W_{net}$$

At constant temperature and pressure,

$$dG = -W_{net}$$

$$\text{or } -(\Delta G)_{T,P} = W_{\text{net}}$$

Thus, the decrease in free energy is equal to the amount of maximum work which a system can do at constant temperature and pressure over and above the mechanical work.

Variation of free energy with Pressure at constant temperature

$$\text{We know that } dG = VdP - SdT - W_{\text{net}}$$

If the system does only mechanical work, then $W_{\text{net}} = 0$

$$dG = VdP - SdT$$

At constant temperature, $dT = 0$

$$(dG)_T = VdP$$

The total change in free energy can be calculated by changing the state of the system from initial(i) to final(f)

$$\int_i^f dG = \int_i^f VdP$$

For ideal gases, $PV = nRT$ Or $V = \frac{nRT}{P}$

$$\Delta G = nRT \int_i^f \frac{dP}{P} = nRT \ln \frac{P_f}{P_i}$$

$$\text{Thus, } \Delta G = 2.303 nRT \log \frac{P_f}{P_i}$$

$$\Delta G = 2.303 nRT \log \frac{V_i}{V_f}$$

With the help of above equation, we can calculate the free energy change for expansion or compression.

Variation of free energy with temperature at constant pressure

Gibbs- Helmholtz equation

For a system performing only mechanical work, $dG = VdP - SdT$

In an isobaric change, pressure is a constant and $dP = 0$

$$\text{i.e., } (dG)_P = - SdT$$

$$\text{or } - \left(\frac{\partial G}{\partial T} \right)_P = S$$

$$\text{As } G = H - TS$$

$$G = H + T \left(\frac{\partial G}{\partial T} \right)_P \text{ or } H = G - T \left(\frac{\partial G}{\partial T} \right)_P$$

$$\text{For a change, } \Delta G = \Delta H + T \left[\frac{\partial (\Delta G)}{\partial T} \right]_P$$

This is known as Gibbs – Helmholtz equation and is applicable to all processes occurring at constant pressure.

$$\text{We have } H = G - T \left(\frac{\partial G}{\partial T} \right)_P$$

On dividing the above equation by T^2 , we get

$$\frac{G - T \left(\frac{\partial G}{\partial T} \right)_P}{T^2} = H/T^2$$

$$\begin{aligned} \text{Now } \frac{\partial}{\partial T} (G/T) &= \frac{\partial}{\partial T} \left(G \times \frac{1}{T} \right) = G \frac{\partial}{\partial T} \left(\frac{1}{T} \right) + \left(\frac{1}{T} \right) \left(\frac{\partial G}{\partial T} \right) \\ &= G \left(-\frac{1}{T^2} \right) + \left(\frac{1}{T} \right) \left(\frac{\partial G}{\partial T} \right) = \frac{-G + T \left(\frac{\partial G}{\partial T} \right)}{T^2} \end{aligned}$$

$$\text{i.e., } \frac{G - T \left(\frac{\partial G}{\partial T} \right)}{T^2} = - \frac{\partial}{\partial T} (G/T)$$

$$\frac{\partial}{\partial T} (G/T)_P = - H/T^2$$

$$\text{Or } \left[\frac{\partial \left(\frac{G}{T} \right)}{\partial T} \right]_P = - \frac{H}{T^2}$$

$$\text{For a change, } \left[\frac{\partial \left(\frac{\Delta G}{T} \right)}{\partial T} \right]_P = - \frac{\Delta H}{T^2}$$

These are other forms of Gibbs- Helmholtz equation

Heat of reaction

A chemical reaction or a physical process is always accompanied by absorption or release of heat by the system. The heat (enthalpy) of reaction is the amount of heat absorbed or evolved in a particular reaction, when the number of moles of reactants represented by the balanced chemical equation change completely into products. It is the difference between the heat content or enthalpies of the products and reactants. It is represented as

$$\Delta H = \text{Enthalpy of products} - \text{Enthalpy of reactants}$$

When the heat of products is more than that of reactants, ΔH is positive. Heat will be absorbed during such a reaction, and the reaction will be endothermic. On the other hand, if the heat of reactants is more than that of products, ΔH is negative. Heat will be evolved in such a reaction and the reaction will be exothermic.

Heats of reaction can be of different types.

Heat (enthalpy) of formation: It is the enthalpy change of a system when one mole of a compound is formed from its constituent elements. It is represented as ΔH_f .

Standard heat of formation is the enthalpy change when one mole of a compound is from its constituent elements, at 298K and one atmosphere pressure. It is represented as ΔH_f^0 .

Heat (enthalpy) of combustion

It is the enthalpy change when one mole of the substance is completely burnt in excess of air. Since heat is always evolved during a combustion reaction heat of combustion is always negative.

Kirchoff's law – The temperature dependence of heat of reaction

For a reaction $X \rightarrow Y$,

$$\Delta H = H_Y - H_X$$

$$\begin{aligned}\frac{d}{dT}(\Delta H)_p &= \left[\frac{\partial H_Y}{\partial T}\right]_p - \left[\frac{\partial H_X}{\partial T}\right]_p \\ &= (C_p)_Y - (C_p)_X = \Delta C_p\end{aligned}$$

This is the well-known Kirchoff's equation and can be used to calculate ΔH at a temperature, if ΔH at some other temperature is known.

$$\frac{d}{dT}(\Delta H)_p = \Delta C_p \quad \text{or} \quad d(\Delta H)_p = \Delta C_p dT$$

Integrating the above equation between two temperatures,

$$\Delta H_2 - \Delta H_1 = \Delta C_p (T_2 - T_1)$$

Free energy and equilibrium constant

From the definition of free energy,

$$G = H - TS = E + PV - TS$$

We obtain

$$dG = dE + PdV + VdP - TdS - SdT$$

But for a situation in which only pressure-volume work can be done, $dE = dq - PdV$, so

$$dG = dq + VdP - TdS - SdT$$

Equating TdS and dq then gives us

$$dG = VdP - SdT$$

For a pressure change at constant temperature

$$dG = VdP$$

With one mole of an ideal gas, we obtain

$$dG_M = \frac{RT}{P} dP$$

On integrating this expression, taking as one limit of pressure $P^0 = 1$ atm, the standard pressure. Then the corresponding limit for G_M will be G_M^0 , the standard free energy of one mole of the ideal gas. We get

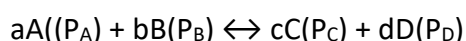
$$\int_{G_M^0}^{G_M} dG = \int_{P^0}^P \frac{RT}{P} dP$$

$$G_M - G_M^0 = RT \ln \frac{P}{P^0} = RT \ln P \rightarrow (1)$$

Where G_M is the molal free energy at any pressure P (in atm) and G_M^0 the standard free energy. If, instead of 1 mole, n moles are considered, we get

$$nG_M = nG_M^0 + nRT \ln P$$

Now, for a general reaction between ideal gases,



where P_A , P_B , P_C and P_D are the pressures of reactants and products.

$$\text{We have } \Delta G_M = cG_M(C) + dG_M(D) - aG_M(A) - bG_M(B)$$

Use of equation (1) gives

$$\Delta G_M = [cG_M^0(C) + dG_M^0(D) - aG_M^0(A) - bG_M^0(D)] + cRT \ln P_C + dRT \ln P_D - aRT \ln P_A - bRT \ln P_B$$

The bracketed terms are equal to ΔG^0 , and the remaining terms can be combined to give

$$\Delta G_M = \Delta G_M^0 + RT \ln \frac{(P_C)^c (P_D)^d}{(P_A)^a (P_B)^b} = \Delta G_M^0 + RT \ln Q$$

When reactants and products are in equilibrium with each other, then $\Delta G_M = 0$, since the initial and final states are in equilibrium, and

$$0 = \Delta G_M^0 + RT \ln Q_{eq}.$$

Because the pressures are those that exist at equilibrium, the reaction quotient at equilibrium is equal to the equilibrium constant K, so

$$\Delta G_M^0 = - RT \ln K \rightarrow (2)$$

Equation (2) gives the quantitative relationship between the standard free energy change and the equilibrium constant.

Chemical potential or partial molar free energy

When a substance is a component of a mixture it makes a contribution to the properties of the mixture that depends on the composition. A partial molar property is the contribution (per mole) that a substance makes to an overall property of a mixture. The concept of partial molar quantity can be applied to state functions like Gibbs free energy. Partial molar free energy or Chemical potential, μ , is the increase in the Gibbs free energy G per mole that results when a small amount dn_i moles of the i th component are added to the system keeping temperature, pressure and all other compositions constant.'

$$\text{Thus, } \mu_i = \left[\frac{\partial G}{\partial n_i} \right]_{T,P}$$

The chemical potential of a gas in a mixture of gases can be expressed as

$\mu_i = \mu_i^0 + RT \ln \frac{P_i}{P^0}$ where μ_i^0 is the standard chemical potential or the standard molar Gibbs energy of the gas i. The chemical potential is an indication of the potential of a substance to be active physically or chemically: the higher the partial pressure, the more active is the species. Thus the chemical potential represents the tendency of the substance to change either physically or chemically when it is at 1 atm (significance of the term μ_i^0). Thus, if the partial pressure of a gas in a mixture changes from P_i to P_f , the change in its chemical potential is

$$\begin{aligned} \Delta \mu_i &= \left[\mu_i^0 + RT \ln \frac{P_f}{P^0} \right] - \left[\mu_i^0 + RT \ln \frac{P_i}{P^0} \right] \\ &= RT \left[\ln \frac{P_f}{P^0} - \ln \frac{P_i}{P^0} \right] = RT \left[\ln \frac{P_f}{P_i} \right] \end{aligned}$$

Fugacity

When thermodynamic equations for ideal gases are applied to real substances, their generality and simplicity are not reproduced. A new thermodynamic quantity called fugacity is introduced so that non-ideal systems could be handled by free energy equations for ideal systems. Lewis stated that each substance in a particular state has a definite tendency to escape from that state and this tendency can be measured by a quantity called fugacity(f). The free energy of the substance per mole is related to fugacity by the expression

$$G = RT \ln f + B$$

B is a constant dependent on temperature and nature of substance. In the standard state:

$$G^0 = RT \ln f^0 + B$$

The difference in free energy at any state and in standard state is given by

$$G - G^0 = RT \ln \frac{f}{f^0}$$

$$\text{Or } G = G^0 + RT \ln \frac{f}{f^0} = G^0 + RT \ln a$$

The ratio of fugacity at any state to fugacity in standard state is called activity(a). In the standard state $G = G^0$, therefore, $RT \ln a = 0$ or $a = 1$. This implies that in standard state activity is unity. The change in free energy from one state (G_1, a_1) to another (G_2, a_2) can be given by

$$\begin{aligned} \Delta G &= G_2 - G_1 = (G^0 + RT \ln a_2) - (G^0 + RT \ln a_1) \\ &= RT \ln \frac{a_2}{a_1} \quad (\text{For 1 mole of substance}) \\ &= nRT \ln \frac{a_2}{a_1} \quad (\text{For } n \text{ moles}) \end{aligned}$$

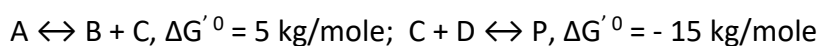
When pressure and concentration of gases and solutions are substituted by activities, exact free energy calculation becomes possible without breakdown of thermodynamic relations.

Thermodynamics of biochemical reactions

Coupling of reactions

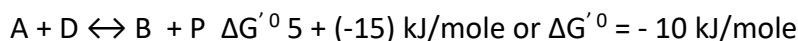
Many individual biochemical processes are nonspontaneous by themselves and have positive ΔG values. However, by adding a second, spontaneous reaction with a negative ΔG , the sum of the two reactions may result in an overall negative ΔG and, therefore, become overall spontaneous. This practice is referred to as coupling the reactions.

For example, consider two reactions with their associated free energy changes under standard conditions:



The first reaction is not spontaneous under standard conditions, but the second is; B and C will not form spontaneously from A under standard conditions.

Now suppose that, for biological reasons, it would be desirable to drive the conversion of A over to B. This can be done by combining or coupling the unfavourable first reaction with the favourable second reaction, thereby taking advantage of the tendency of C to react with D. The net result is



The coupled reaction is now spontaneous under standard conditions. Thus, feeding in A and D leads spontaneously to the formation of B and P; The coupling of unfavourable reactions with favourable ones appears widely throughout biochemistry.

Phase Rule

A phase is defined as any homogeneous and physically distinct part of a system bounded by a surface and is mechanically separable from other parts of the system.

A phase may be gaseous, liquid or solid. There must be a definite boundary, known as interface, between any two phases and there exists a dynamic equilibrium in between them.

A system consisting of only one phase is said to be homogeneous.

A mixture of two immiscible liquids such as water and benzene, will exist in two distinct liquid phases and in addition there will be a vapour phase. Thus there will be three phases each separated from the other by a well-defined interface.

A system consisting of more than one phase is said to be heterogeneous.

The homogeneous reversible reactions can be studied using the law of mass action.

The phase rule given by Williard Gibbs is used for studying heterogeneous reversible reactions.

Statement : Phase rule states that “ If the equilibrium between any number of phases is influenced only by temperature, pressure and concentration, then the number of degrees of freedom (F) of the system is related to the number of components (C) and number of phases (P) by the following phase rule equation :

$$F = C - P + 2$$

Explanation of terms

1) Phase (P)

A phase is defined as a homogeneous, physically distinct and mechanically separable portion of system, which is separated from other such parts of the system by definite boundary surfaces.

Example :

1. Liquid phase :

The number of liquid phase depends on the number of liquids present and their miscibility.

i) If two liquids are immiscible, they will form two separate liquid phases.

Example : benzene and water

ii) If two liquids are miscible they will form one liquid phase only.

Example : alcohol and water

2. Solid phase

Each solid forms a separate phase. The number of solid phase depends on the number of solids present in it.

Example : Many forms of sulphur can exist together, but these are all separate phases.

3. Gaseous phase

Since a gaseous mixture are thoroughly miscible in all proportions, it will form one phase only.

Example : a mixture of N_2 and H_2 forms one phase only.

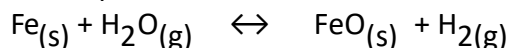
4. A solution of a substance in a solvent consists of one phase only, e.g. glucose solution.

5. A heterogeneous mixture like:



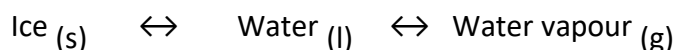
consists of three phases (i.e., two solids and one gaseous).

In the equilibrium reaction,



There are two solid phases, Fe and FeO and one gaseous phase consisting of $\text{H}_2\text{O}(\text{g})$ and $\text{H}_2(\text{g})$. Thus three phases exists in equilibrium.

6. At freezing point, water consists of three phases:



II) Component (C)

Component is defined as “the smallest number of independently variable constituents, by means of which the composition of each phase can be expressed in the form of a chemical equation”.

Example :

i) In the water system,



The chemical component of all the three phases is H_2O and therefore it is one component system.

ii) Sulphur exists in four phases namely rhombic, monoclinic, liquid and vapour, but the chemical composition of all phases is S. Thus is a one component system.

iii) A system of saturated solution of NaCl consists of solid salt, salt solution and water vapour. The chemical composition of all the three phases can be expressed in terms of NaCl and H_2O . Therefore it is a two component system.

iv) In the thermal decomposition of CaCO_3 ,



The composition of each of the three phases can be expressed in terms of at least any two of the independent variable constituents, CaCO_3 , CaO and CO_2 . Suppose CaCO_3 and CaO are chosen as the two components, then the composition of different phases is represented as follows:

Phase : $\text{CaCO}_3 = \text{CaCO}_3 + 0\text{CaO}$

Phase : $\text{CaO} = 0\text{CaCO}_3 + \text{CaO}$

Phase : $\text{CO}_2 = \text{CaCO}_3 - \text{CaO}$

Thus, it is a two component system..

v) In the equilibrium, $\text{Fe}(\text{s}) + \text{H}_2\text{O}(\text{g}) \leftrightarrow \text{FeO} (\text{s}) + \text{H}_2 (\text{g})$,

the minimum components required to express the composition of each phase is three.

Thus it is a three component system.

III) Degree of freedom (F)

Degree of freedom is defined as the minimum number of independent variable factors such as temperature, pressure and concentration of the phases, which must be fixed in order to define the condition of a system completely.

A system having 1,2,3 or 0 degrees of freedom is called univariant, bivariant, trivariant and nonvariant respectively.

Example :

i) Consider the water system,



The three phases can be in equilibrium only at particular temperature and pressure. Therefore, when all the three phases are present in equilibrium, then no condition need to be specified. The system is therefore zero variant or invariant or has no degree of freedom.

In this system if pressure or temperature is altered, three phases will not remain in equilibrium and one of the phases disappears.

ii) Consider a system consisting of water in contact with its vapour,

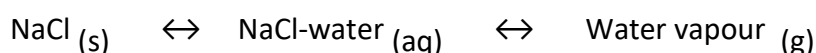


To define this system completely, we must state either the temperature or pressure. Thus degree of freedom is one and the system is univariant.

iii) For a system consisting of water vapour phase only, we must state the values of both the temperature and pressure in order to define the system completely. Hence the system is bivariant or has two degrees of freedom.

iv) For a gaseous mixture of N_2 and H_2 , we must state both the pressure and temperature, because if pressure and temperature are fixed, the volume automatically becomes definite. Hence, for a gaseous system, two factors must be stated in order to define it completely and thus, it has two degrees of freedom or bivariant system.

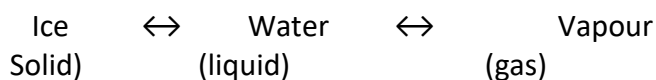
v) Consider a system consisting of



we must state either the temperature or pressure, because the saturation solubility is fixed at a particular temperature or pressure. Hence the system is univariant.

Applications of Phase rule to one component system (water system)

The water system is a one component system



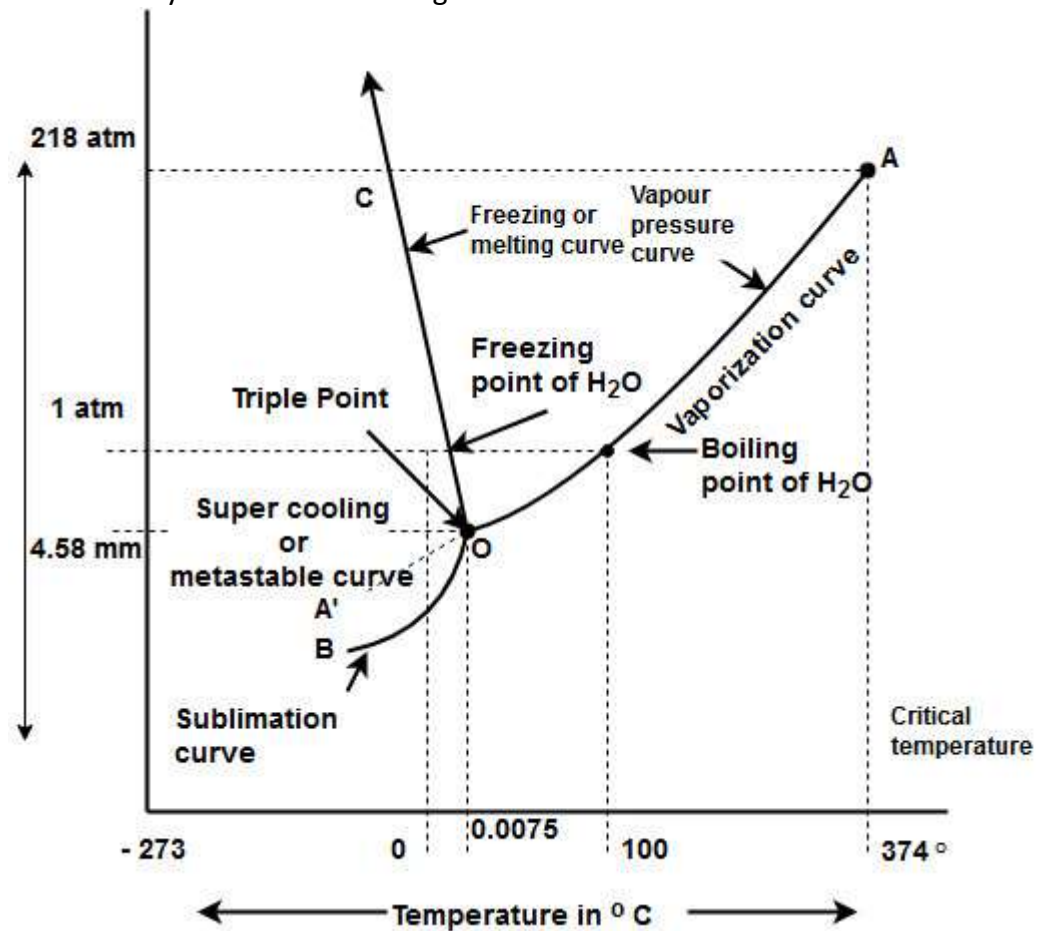
Since water exists in three possible phases such as solid, liquid and vapour, there are three forms of equilibria :

Liquid – vapour, solid vapour and solid liquid

i.e.,	Liquid	Vapour
	Solid	Vapour
	Solid	Liquid

Each equilibrium involves two phases. The nature of these phases which exist in equilibrium at any time depends on the conditions of temperature and pressure. These conditions have been determined and summarized in the pressure-temperature diagram called phase diagrams in which pressure is treated as independent variable and is plotted along y – axis whereas temperature is plotted along x- axis.

The phase diagram for the water system is shown in Fig. shown below:



The phase diagram consists of

1. Curves : There are three curves OA, OB and OC.
2. Areas : Three curves OA , OB and OC divide the diagram into three areas AOB, AOC and BOC.
3. Triple point : The above three curves meet at the point O and is known as triple point.
4. Metastable equilibrium : The curve OA' represents the metastable equilibrium.

1) Curve OA

The curve OA is called vapourisation curve, it represents the equilibrium between water and vapour. At any point on the curve the following equilibrium will exist.



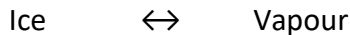
The degree of freedom of the system is one, i.e, univariant. Thus applying phase rule equation,

$$F = C - P + 2 = 1 - 2 + 2 ; F = 1$$

This equilibrium (i.e., line OA) will extend upto the critical temperature (374°C). Beyond the critical temperature the equilibrium will disappear only water vapour will exist.

2) Curve OB

The curve OB is called sublimation curve of ice, it represents the equilibrium between ice and vapour. At any point on the curve the following equilibrium will exist.



The degree of freedom of the system is one, i.e., univariant.

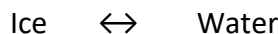
This is predicted by the phase rule.

$$F = C - P + 2 ; F = 1 - 2 + 2 ; F = 1$$

This equilibrium line will extend upto the absolute zero (– 273°C) where no vapour can be present and only ice will exist.

3) Curve OC

The curve OC is called melting point curve of ice, it represents the equilibrium between ice and water. At any point on the curve the following equilibrium will exist.



The curve OC is slightly inclined towards pressure axis. This shows that melting point of ice decreases with increase of pressure.

The degree of freedom of the system is one. i.e., univariant.

iv) Triple point (Point 'O')

At triple point all the three phases namely ice, water and vapour coexist. Thus the value of P is 3. Applying phase rule equation, the degree of freedom at this point is zero. It means that three phases can coexist in equilibrium only at a definite temperature and pressure. The values are 0.0075 °C and 4.58 mm respectively. At this triple point, neither pressure nor temperature can be altered even slightly without causing the disappearance of one of the phases. The triple point is not the same as the ordinary melting point of ice (i.e., 0 °C). Its value has been increased due to the fact that 0 °C is the melting point of ice at 760mm of mercury and a decrease of 4.58 mm will rise the melting point to 0.0075°C.

v) Curve OB (Metastable equilibrium)

The curve OA' is called vapour pressure curve of the super-cool water or metastable equilibrium. Where the following equilibrium will exist.



Sometimes water can be cooled below 0°C without the formation of ice, this water is called super-cooled water. Supercooled water is unstable and it can be converted into solid by 'seeding' or by slight disturbance.

vi) Areas

Area AOC, BOC, AOB represents water, ice and vapour respectively. In order to define the system at any point in the areas, it is essential to specify both temperature and pressure.

The degree of freedom of the system is two. i.e., Bivariant.

This is predicted by the phase rule

$$F = C - P + 2; F = 1 - 1 + 2; F = 2$$

Two component system

Reduced phase rule or condensed phase rule.

We know the phase-rule equation, $F = C - P + 2$ (1)

For a two component system, $C = 2$ and hence the above equation becomes,

$$F = 2 - P + 2 = 4 - P \text{ (2)}$$

The minimum number of phases in any system at equilibrium is one. It is clear from the equation (2), the maximum number of degree of freedom is three.

Thus, three variables – pressure, temperature and composition of one of the components must be specified to describe the system. This will lead to three dimensional figures which cannot be conveniently represented on a paper. To make this simple, one of the three variables is kept constant.

In solid-liquid equilibrium of an alloy, practically there is no gaseous phase and the pressure will not have much influence. In the case of solid-liquid equilibrium, the experiments are generally carried out at constant pressure.

Thus the system in which only the solid and liquid phases are considered and the gas phase is ignored is called a condensed system. This reduces the degree of freedom of the system by one. The phase rule equation is then written as

$$F' = C - P + 1 \text{ (3)}$$

This equation is called reduced phase rule or condensed phase rule.

For a two component system the phase rule equation is written as

$$F' = C - P + 1 = 2 - P + 1 = 3 - P \text{ (4)}$$

The above equation is known as the reduced (condensed) form of phase rule for two component system.

Classification of two component system

The two component systems are classified into the following three types :

- i) Simple eutectic formation
- ii) a) Formation of compound with congruent melting point. b) Formation of compound with incongruent melting point.
- iii) iii) Formation of solid solution.

i) Simple eutectic formation :

A system with two substances which are completely miscible in the liquid state, but completely immiscible in the solid state is known as eutectic system. In this system the substances do not react chemically. Among the mixtures of different proportions of two substances, the mixture which has the lowest melting point is known as the eutectic mixture. The temperature and composition corresponding to the point eutectic point is called eutectic temperature and eutectic composition respectively.

ii) a) Formation of compound with congruent melting point : In this type of binary alloy system the two substances form one or more compounds with definite proportions. Of the compounds, a compound is said to possess congruent melting point, if it melts exactly at a constant temperature into liquid, having the same composition as that of the solid.

b) Formation of compound with incongruent melting point : Of the above compounds, a compound is said to possess incongruent melting point, if it decomposes completely at a temperature below its melting point yielding a new solid phase with a composition different from that of the original.

iii) Formation of solid solution : In this type, when two substances, especially metals, are completely miscible in both the solid and liquid states, they form solid solutions where

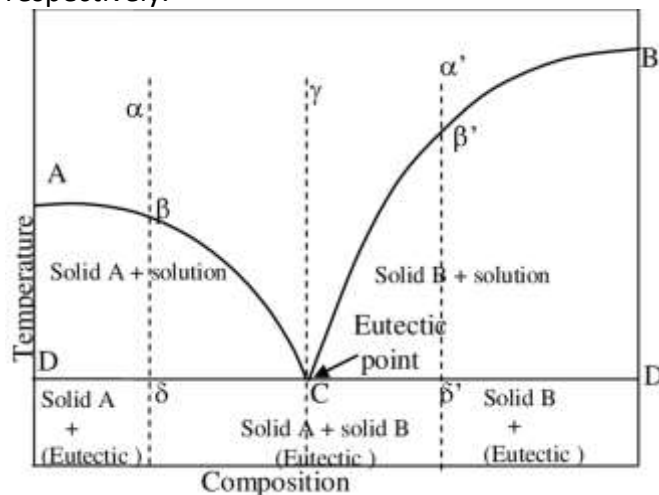
mixing takes place in the atomic levels. A solid solution can be formed only when the difference between the atomic radius of two metals is not greater than 15%.

Simple eutectic systems

The general phase diagram for binary alloy systems is shown in Fig 4.2 . Here the pressure does not have the considerable effect. Hence, the other two variables viz, temperature and compositions are taken into account.

Components A and B.

When small quantities of B are added to A gradually, the melting point of A falls along the curve AC. In the same way when small quantities of A are added to B gradually, the melting point B falls along the curve BC. Hence, AC and BC are the freezing point curves of A and B respectively.



The Simple eutectic systems

The curves AC and BC meet at the point C. At this point the three phases solid A, solid B and their solution coexist. The degree of freedom is zero here and the system is therefore invariant. Also only at this point C, the liquid can exist at the lowest temperature. Since the mixture of components A and B of composition corresponding to the point C has the lowest melting point, the point C is called the eutectic point. The temperature and composition corresponding to the point C is called eutectic temperature and eutectic composition respectively.

Consider a liquid mixture of composition represented by a point α cooled at constant pressure. The temperature falls without any change of composition until the point on the curve AC is reached. At this temperature t_1 , the solid A will separate out. The system now consists of two phases and hence monovariant. As cooling continues, the component A keeps on separating out and the solution becomes relatively richer in B. The temperature and the solution composition both change along AC. Thus at the temperature t_1 , solid A is in equilibrium with solution of composition X and at temperature t_2 , it is in equilibrium with solution of composition Y.

It is clear therefore, in the area ACD, solid A is in equilibrium with solutions of varying composition given by the curve AC depending upon the temperature.

When the temperature reaches a point represented by δ , the solid B also begins to separate out. On further cooling the system, solid A and B separate out together in constant ratio so that the composition of the solution remains constant. The temperature also remains constant for some time. When the liquid solution has been completely solidified

and the system consists only of a mixture of solid A and B, it becomes monovariant. Further cooling will result in the fall of temperature below the line DD into the area in which only the two solids coexist as shown.

In the same way, if the composition of liquid mixture is on the right of the eutectic point C, as represented by point α' , similar series of changes will be obtained on cooling.

Merits of the Phase rule:

1. It is applicable to both physical and chemical equilibria.
2. It requires no information regarding molecular/micro-structure, since it is applicable to macroscopic systems.
3. It is a convenient method of classifying equilibrium states in terms of phases, components and degrees of freedom.
4. It helps us to predict the behaviour of a system, under different sets of variables.
5. It indicates that different systems with same degree of freedom behave similarly.
6. It helps in deciding whether under a given set of conditions :
 - a) various substances would exist together in equilibrium or
 - b) some of the substances present would be interconverted or
 - c) some of the substances present would be eliminated.

Limitations of Phase rule:

1. It can be applied only for system in equilibrium. Consequently, it is of little value in case of very slow equilibrium state attaining system.
2. It applies only to a single equilibrium system; and provide no information regarding any other possible equilibria in the system.
3. It requires at most care in deciding the number of phases existing in an equilibrium state, since it considers only the number of phases, rather than their amounts. Thus even if a trace of phase is present, it accounts towards the total number of phases.
4. It conditions that all phases of the system must be present simultaneously under the identical conditions of temperature and pressure.
5. It conditions that solid and liquid phases must not be in finely-divided state; otherwise deviations occurs.

